

Elementary Topology

Generalizing the concept of continuity to general sets of points

LXJ

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Motivation: from calculus to topology

Recall (continuity in $\mathbb{R}^N$)

A map $f: \mathbb{R}^N \rightarrow \mathbb{R}^M$ ($N = M = 1$, for e.g.) is *continuous* if

$$\forall x_0 \in \mathbb{R}^N : \quad \lim_{x \rightarrow x_0} f(x) = f(x_0),$$

which is equivalent to the following description:

for every open set $U \subset \mathbb{R}^M$, $f^{-1}(U)$ is open in $\mathbb{R}^N$.

Key

To define continuity, we need the notion of **open sets**.

Topological Spaces

Topological structure

Definition (Topological structure)

Let S be a non-empty set. A *topological structure* $\mathcal{T}$ on S is a collection of subsets of S such that

1. $\emptyset, S \in \mathcal{T}$;
2. if $U_\alpha \in \mathcal{T}$ for $\alpha \in I$ (I being some index set), then $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}$;
3. if $U_1, \dots, U_n \in \mathcal{T}$, then $\bigcap_{i=1}^n U_i \in \mathcal{T}$.

The pair $(S, \mathcal{T})$ is called a *topological space*.

The elements of $\mathcal{T}$ are called *open sets*.

Definition (Closed set)

The complement of an open set is called a *closed set*.

Continuous Maps

Continuous maps between topological spaces

Definition (Continuous maps between topological spaces)

A map $f: X \rightarrow Y$ between topological spaces X and Y is *continuous* if for every open set $V \subset Y$, the preimage $f^{-1}(V)$ is an open set of X .

Proposition (Composition of continuous maps)

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are continuous maps, then $g \circ f: X \rightarrow Z$ is continuous.

Proposition (Continuity defined via closed sets)

A map $f: X \rightarrow Y$ is continuous $\iff$ the preimage of every closed set is closed.

Open maps, closed maps, homeomorphisms

Definition (Open maps and closed maps)

A map $f: X \rightarrow Y$ is called an *open map* if the image $f(U) \subset Y$ is open for every open set $U \subset X$;

$f: X \rightarrow Y$ is called *closed* if $f(V) \subset Y$ is closed for every closed $V \subset X$.

Definition (Homeomorphism (同胚))

A bijection (双射) $f: X \rightarrow Y$ is a *homeomorphism* if both f and f^{-1} are continuous. Topological spaces X and Y are then said to be *homeomorphic*.

Subspace Topology

Subspace topology

Definition (Subspace topology)

Let $(S, \mathcal{T})$ be a topological space and $A \subset S$ be a subset. Then

$$\mathcal{T}_A := \{A \cap U \mid U \in \mathcal{T}\}$$

is the *subspace topology* on A .

Exercise

Verify that $\mathcal{T}_A$ defines a topological structure on A .

Exercise

Define the natural inclusion $\iota : A \hookrightarrow S$. Show that $(A, \mathcal{T}_A)$ is a subspace structure $\iff \iota$ is continuous.

Bases

Basis of a topology

Definition (Basis of a topology)

A subcollection $\mathcal{B} \subset \mathcal{T}$ is a *basis* for $\mathcal{T}$ if $\forall U \in \mathcal{T}$ and $p \in U$, $\exists B \in \mathcal{B}$ such that

$$p \in B \subset U.$$

We also say that $\mathcal{B}$ *generates* the topology $\mathcal{T}$, or that $\mathcal{B}$ is a basis for the topological space S .

Example

$\mathcal{B} = \{ B(p, r) \mid p \in \mathbb{R}^n, r > 0 \}$ is a basis for the standard topology on $\mathbb{R}^n$.

Characterizations of a basis

Proposition

$\mathcal{B}$ is a basis for $\mathcal{T}$ iff every $U \in \mathcal{T}$ can be written as

$$U = \bigcup_{\text{some } B \in \mathcal{B}} B.$$

Proposition (Criterion)

A collection $\mathcal{B}$ of subsets of a set S is a basis for a topology iff

1. $S = \bigcup_{B \in \mathcal{B}} B$;
2. $\forall B_1, B_2 \in \mathcal{B}$ and $p \in B_1 \cap B_2$ (if it exists), $\exists B \in \mathcal{B}$ such that $p \in B \subset B_1 \cap B_2$.

Proposition

Let $\mathcal{B} = \{B_\alpha\}$ be a basis for the topological space S , and let A be a subspace of S . Then $\{B_\alpha \cap A\}$ is a basis for A .

First and Second Countability

Proposition

Let $\mathcal{B}_{\text{rat}} := \{ B(p, r) \mid p \in \mathbb{Q}^n, r \in \mathbb{Q}^+ \setminus \{0\} \}$. Then $\mathcal{B}_{\text{rat}}$ is a basis for $\mathbb{R}^n$.

Definition (Second countable)

A topological space is said to be *second countable* if it has a *countable basis*.

First countability

Definition (Neighborhood basis)

Let S be a topological space and $p \in S$. A *neighborhood basis* at p is a collection $\mathcal{B} = \{B_\alpha\}$ of neighborhoods of p such that for every neighborhood U of p , $\exists B_\alpha \in \mathcal{B}$ with

$$p \in B_\alpha \subset U.$$

Definition (First countable)

S is said to be *first countable* if it has a countable neighborhood basis at every point $p \in S$.

Separation Axioms

Hausdorff and normal spaces

Definition (Hausdorff topological spaces)

A topological space S is called a *Hausdorff space* if $\forall$ points $x, y \in S$, $\exists$ two *disjoint open sets* U, V such that $x \in U, y \in V$.

Definition (Normal spaces)

A Hausdorff space is called *normal* if, given any two disjoint closed sets F, G , $\exists$ disjoint open sets U, V such that

$$F \subset U \quad \& \quad G \subset V.$$

Proposition

Any one-point set in a Hausdorff space is closed.

Exercise

Any subspace of a Hausdorff space is also Hausdorff.

Examples

Example

$\mathbb{R}^n$ is a Hausdorff topological space.

Example (The Zariski topology)

Let $S = \mathbb{C}^n$ be the n -dimensional vector space over $\mathbb{C}$, endowed with the *Zariski topology*. Every open set U is of the form $S \setminus Z(I)$, where I is a family of polynomials in n variables and $Z(I)$ is the solution set of this family.

Any two such open sets intersect:

$$U \cap V = S \setminus (Z(I) \cup Z(J))$$

is nonempty, since $Z(I) \cup Z(J)$ — being the solution set of I or J — cannot be the whole space. Hence the Zariski topology is *not* Hausdorff.

Product Topology

Product topology

The Cartesian product $A \times B = \{(a, b) \mid a \in A, b \in B\}$.

Definition (Product topology)

Given two topological spaces X and Y , construct the collection $\mathcal{B}$ consisting of all sets of the form $U \times V$ (U, V open sets in X and Y). $\mathcal{B}$ generates a topology on $X \times Y$, called the *product topology*.

Proposition

Let $\{U_i\}$ and $\{V_j\}$ be bases for X and Y respectively. Then $\{U_i \times V_j\}$ is a basis for $X \times Y$.

Proposition

The product of two second countable topological spaces is second countable.

Exercise

The product of two Hausdorff topological spaces is Hausdorff.

Compactness

Covers and compact spaces

Definition (Cover, subcover)

Let S be a topological space. A collection of open sets $\{U_\alpha\}_{\alpha \in A}$ of S is called a *cover* (or *open cover*) of S if $S = \bigcup_{\alpha} U_\alpha$. If a subcollection $\{U_\alpha\}_{\alpha \in B \subset A}$ of $\{U_\alpha\}_{\alpha \in A}$ still covers S , the subcollection is called a *subcover* of S .

Definition (Compact topological space)

A topological space S is called *compact* if every open cover of S has a *finite subcover*.

Compactness: basic properties

Exercise

A closed subset of a compact topological space is compact.

Exercise

Every compact subset K of a Hausdorff space S is closed.

Proposition

The image of a compact set under a continuous map is compact.

Maps from compact to Hausdorff spaces

Proposition

A continuous map from a compact topological space X to a Hausdorff topological space Y is a closed map.

Corollary

A continuous bijection f from a compact topological space X to a Hausdorff space Y is a homeomorphism.

Exercise

Prove the corollary above.

Connectedness

Connected spaces and connected components

Definition (Connected topological spaces)

A topological space S is called *connected* if there do *not* exist two nonempty disjoint open subsets U and V such that $S = U \cup V$. A subset $A \subset S$ is called *connected* if it is connected as a topological subspace.

Proposition

The image of a connected topological space X under a continuous map $f : X \rightarrow Y$ is connected.

Definition (Connected components)

Let x be a point of S . The set C_x — the union of all connected subsets of S containing x — is called the *connected component* of S containing x .

Closure

Closure

Definition (Closure)

The *closure* of A in S , denoted by $\bar{A}$, is defined as the intersection of all closed sets containing A (the smallest closed set containing A).

Proposition (Local characterization of the closure)

Let $A \subset S$. A point $p \in S$ lies in $\bar{A}$ iff every neighborhood of p contains at least one point of A .

Proposition

A subset $A \subset S$ is closed iff $A = \bar{A}$.

Proposition

If $A \subset B$, then $\bar{A} \subset \bar{B}$.

Further Exercises

Exercises and further reading

Exercise

What is a paracompact topological space?

Exercise

Read the part on the quotient topology.